A Probabilistic Framework for Combo Assembly in Constructed Card Games: Hypergeometric Models, Deck Composition, and Two-Player Race Dynamics

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Abstract

We develop a probabilistic model for combo assembly in constructed card games—specifically Magic: The Gathering—where a player must draw a specific subset of cards from a shuffled deck to execute a winning combination. The model rests on the multivariate hypergeometric distribution, parameterized by deck size, card-type composition, mulligan policy, and additional draw effects. We derive closed-form expressions for the probability of assembling a combo of N cards by turn T, decomposed across card types (lands, creatures, enchantments, instants, sorceries). We then extend the framework to a two-player race: given that Player 1 requires n_1 combo pieces and Player 2 requires $n_2 > n_1$ pieces, we characterize win probabilities as a function of turn progression. A further extension incorporates multiple independent combo paths per player, showing that redundancy in win conditions can offset the disadvantage of larger combo requirements. Monte Carlo simulations in R illustrate all theoretical results, with comparative statics on deck composition, combo size, and the number of independent combo sets.

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1 Introduction

Competitive Magic: The Gathering revolves around deck construction and in-game sequencing. A central strategic concern is *combo assembly*: the probability that a player draws a specific set of cards—the "combo pieces"—needed to execute a winning combination within a relevant time horizon.

Deck-building decisions directly affect this probability. A 60-card deck with 24 lands, 20 creatures, and 16 noncreature spells presents a very different draw distribution than a 60-card deck with 20 lands, 8 creatures, 20 instants, and 12 enchantments. The composition of the deck determines the multivariate hypergeometric law governing draws, and therefore the likelihood of assembling any given combo.

This paper formalizes the problem. Section 2 introduces notation and the probability space. Section 3 derives the core result: the probability of assembling an N-card combo by turn T under the multivariate hypergeometric distribution, with extensions for mulligans (Section 4) and extra draw effects (Section 5). Section 6 analyzes how deck composition shifts the probability surface. Section 7 introduces the two-player race model, and Section 8 extends it to multiple independent combos. Section 9 presents Monte Carlo simulations and charts. Section 11 concludes.

2 Setup and Notation

Definition 1 (Deck)**.** *A* deck $\mathcal{D}$ *consists of* X *cards. In standard constructed Magic,* $X = 60$*, though we keep the notation general.*

Definition 2 (Card-Type Partition)**.** *The deck is partitioned into* M *disjoint card types,*

$$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \cdots \cup \mathcal{D}_M, \qquad |\mathcal{D}_i| = K_i, \qquad \sum_{i=1}^{M} K_i = X.$$

A typical partition for MTG is $M = 5$*: lands (*$\mathcal{D}_1$*), creatures (*$\mathcal{D}_2$*), enchantments (*$\mathcal{D}_3$*), instants (*$\mathcal{D}_4$*), and sorceries (*$\mathcal{D}_5$*).*

Definition 3 (Combo)**.** *A* combo $\mathcal{Z}$ *is a target collection requiring at least* n_i *cards from type* $\mathcal{D}_i$*, for* $i = 1, \ldots, M$*. The total combo size is*

$$N = \sum_{i=1}^{M} n_i.$$

Definition 4 (Cards Seen)**.** *Let* $d(T)$ *denote the number of distinct cards a player has drawn (i.e., seen) by the beginning of turn* T*. Under standard rules,*

- *The player draws an opening hand of* h *cards (default* $h = 7$*).*
- *On each subsequent turn* $t = 1, 2, \ldots$*, the player draws 1 card (the player on the play skips the draw on turn 1; we parameterize this).*

Thus, on the play:

$$d(T) = h + \max(T - 1, 0),$$

and on the draw:

$$d(T) = h + T.$$

We write $d(T) = h + T - \mathbf{1}_{play}$*, where* $\mathbf{1}_{play} = 1$ *if on the play, 0 if on the draw.*

Remark 1. *Throughout, we assume* $d(T) \leq X$*, which holds for any realistic game length.*

3 Combo Assembly Probability: Single Player

3.1 The Multivariate Hypergeometric Framework

The deck is shuffled uniformly at random. Drawing d cards without replacement from a deck partitioned into M types with sizes $(K_1, \ldots, K_M)$ yields a draw vector $(Y_1, \ldots, Y_M)$ that follows the *multivariate hypergeometric distribution*:

$$\Pr(Y_1 = y_1, \ldots, Y_M = y_M) = \frac{\prod_{i=1}^{M} \binom{K_i}{y_i}}{\binom{X}{d}},$$

where $\sum_{i=1}^{M} y_i = d$ and $0 \leq y_i \leq K_i$ for all i.

Proposition 1 (Combo Probability)**.** *The probability of assembling combo* $\mathcal{Z} = (n_1, \ldots, n_M)$ *after drawing* d *cards from a deck of size* X *with type sizes* $(K_1, \ldots, K_M)$ *is*

$$P_{\mathcal{Z}}(d) = \Pr\left(\bigcap_{i=1}^{M} \{Y_i \geq n_i\}\right) = \sum_{\substack{y_1,\ldots,y_M \geq 0 \\ y_i \geq n_i \ \forall i \\ \sum y_i = d}} \frac{\prod_{i=1}^{M} \binom{K_i}{y_i}}{\binom{X}{d}}. \tag{1}$$

Proof. Direct application of the multivariate hypergeometric probability mass function, summing over all feasible draw vectors satisfying the combo requirement. □

3.2 The Two-Type Special Case

When the combo requires n specific cards from a pool of K "combo-relevant" cards and the remaining $X - K$ cards are irrelevant, the problem reduces to the univariate hypergeometric:

$$P(d) = \sum_{y=n}^{\min(K,d)} \frac{\binom{K}{y}\binom{X-K}{d-y}}{\binom{X}{d}}. \tag{2}$$

Corollary 1. *If the combo requires exactly* N *specific singleton cards (i.e.,* $K_i = 1$ *for* $i = 1, \ldots, N$ *and* $K_{N+1} = X - N$*), then*

$$P_{\mathcal{Z}}(d) = \frac{\binom{X-N}{d-N}}{\binom{X}{d}} = \prod_{j=0}^{N-1} \frac{d-j}{X-j}. \tag{3}$$

Proof. Each of the N singletons must appear in the d-card draw. The number of favorable outcomes is $\binom{X-N}{d-N}$ (choose the remaining $d-N$ cards from the $X-N$ non-combo cards). Dividing by $\binom{X}{d}$ and simplifying yields the product form. □

Remark 2. *Equation (3) makes clear that the probability increases in d (more cards seen) and decreases in N (larger combo) and X (larger deck), as expected.*

3.3 Cumulative Probability Over Turns

Substituting $d = d(T)$ into (1), we obtain the *turn-by-turn combo assembly probability*:

$$P_{\mathcal{Z}}(T) \equiv P_{\mathcal{Z}}\big(d(T)\big).$$

This is a monotonically nondecreasing function of T, converging to 1 as $d(T) \to X$.

We also define the *first-assembly distribution.* Let τ be the first turn on which the combo is complete:

$$\Pr(\tau = T) = P_{\mathcal{Z}}(T) - P_{\mathcal{Z}}(T-1), \qquad T = 0, 1, 2, \ldots$$

where $P_{\mathcal{Z}}(-1) \equiv 0$ and $T = 0$ corresponds to the opening hand.

4 Mulligan Adjustments

Under the London mulligan rule (current as of 2019), a player may choose to mulligan: they draw a new hand of 7, then put m cards on the bottom of the library, where m is the number of times they have mulliganed. The effective opening hand size after m mulligans is $h = 7 - m$.

Definition 5 (London Mulligan)**.** *After m mulligans, the player has seen 7 cards (from the final mulligan draw), chosen $7-m$ to keep, and placed m on the bottom. The effective hand is $h = 7 - m$ cards, but the player has* information *about 7 cards.*

The London mulligan introduces selection bias: the player keeps the "best" $7-m$ cards from 7. To model this exactly requires specifying a keep/mulligan policy. We consider two approaches:

Approach 1 (Conservative): Treat the effective hand as $h = 7 - m$ cards drawn uniformly at random from X. This ignores the selection effect and provides a lower bound on combo probability.

Approach 2 (Greedy Mulligan): The player mulligans if and only if the 7-card hand does not contain all combo pieces, up to a maximum of $m_{\max}$ mulligans. Let q be the probability that a random 7-card hand does *not* contain all combo pieces:

$$q = 1 - P_{\mathcal{Z}}(7).$$

After m mulligans under this greedy policy, the probability that at least one of the $m+1$ independent hands contained the combo is $1 - q^{m+1}$. However, the player still loses m cards. The effective post-mulligan combo probability (accounting for the bottom cards being unavailable until drawn through the deck) requires a more careful treatment.

For tractability, our simulations implement the greedy policy directly via Monte Carlo.

5 Extra Draw Effects

Many cards in MTG allow drawing additional cards. We model this as an *extra draw rate* $\delta \geq 0$: on each turn after the first, the player draws $1+\delta$ cards on average, where δ captures cantrips, card-draw spells, and similar effects.

The cards-seen function generalizes to

$$d(T) = h + (1+\delta)(T - \mathbf{1}_{\text{play}}).$$

In practice, δ is stochastic and depends on the draw itself (you must draw the draw spells to use them), creating a feedback loop. Our simulations handle this by modeling draw spells as a separate card type. The closed-form expressions use the deterministic approximation above.

6 Deck Composition Effects

The multivariate hypergeometric in (1) depends on the full type-size vector $(K_1, \ldots, K_M)$. Deck composition affects combo probability through two channels:

Channel 1: Combo density. If the combo requires n_i cards from type i, increasing K_i (adding more cards of that type) raises the marginal probability of drawing n_i or more from that type.

Channel 2: Dilution. Adding cards of type $j \neq i$ (with $n_j = 0$) increases X without increasing any relevant K_i, diluting the combo.

Proposition 2 (Monotonicity in Type Size)**.** *Fix all type sizes except K_i. If $n_i \geq 1$, then $P_{\mathcal{Z}}(d)$ is strictly increasing in K_i for $K_i \geq n_i$, holding X fixed (i.e., swapping non-combo cards for combo-type cards).*

Proof. Increasing K_i while decreasing some K_j with $n_j = 0$ raises $\binom{K_i}{y_i}$ for $y_i \geq n_i$ and does not reduce $\binom{K_j}{y_j}$ for $y_j \geq 0$. The denominator $\binom{X}{d}$ is unchanged. Hence the sum increases. □

6.1 Optimal Land Count Under Combo Constraints

A classic deck-building trade-off: lands are necessary to cast spells (mana requirements) but do not contribute to the combo (typically). Let K_L denote the number of lands. If the combo requires n_L lands and $n_S = N - n_L$ spells, the combo probability as a function of K_L (with $K_S = X - K_L$) exhibits an interior maximum when both $n_L > 0$ and $n_S > 0$.

Proposition 3 (Interior Optimum)**.** *If $n_L \geq 1$ and $n_S \geq 1$, there exists a unique $K_L^* \in (n_L, X - n_S)$ that maximizes $P_{\mathcal{Z}}(d)$ for any fixed $d \geq N$.*

The proof follows from the strict log-concavity of the hypergeometric pmf in the type-size parameter, which we omit for brevity. The simulations in Section 9 illustrate this trade-off numerically.

7 Two-Player Combo Race

7.1 Setup

Consider two players in a match. Each player has a deck of size X and seeks to assemble their respective combo. Player i requires n_i combo pieces, with $n_1 \leq n_2$ (Player 1 has a smaller, "faster" combo).

On each turn:

1. Player 1 draws a card.
2. Player 1 checks whether their combo is complete; if so, they win.
3. Player 2 draws a card.
4. Player 2 checks whether their combo is complete; if so, they win.

(We assume Player 1 is on the play, i.e., acts first.)

Definition 6 (Win Probability)**.** *Let τ_i denote the first turn on which Player i assembles their combo. Then*

$$\Pr(\textit{Player 1 wins}) = \Pr(\tau_1 < \tau_2) + \tfrac{1}{2}\Pr(\tau_1 = \tau_2,\ \textit{Player 1 acts first}),$$

where the tie-breaking reflects turn order: if both assemble on the same turn, Player 1 (on the play) wins because they check first.

7.2 Analytical Characterization

Theorem 1 (Win Probability in the Combo Race)**.** *Under the singleton combo model (Corollary 1), the probability that Player 1 wins is*

$$W_1 = \sum_{T=0}^{\infty} \Pr(\tau_1 = T) \cdot \Pr(\tau_2 > T) + \sum_{T=0}^{\infty} \Pr(\tau_1 = T) \cdot \Pr(\tau_2 = T) \cdot \mathbf{1}_{\textit{P1 acts first at } T}, \tag{4}$$

where $\Pr(\tau_i = T)$ is the first-assembly probability for Player i at turn T.

The sums are finite (truncated at $T^* = X - h$). Each term involves differences of hypergeometric CDFs, which are straightforward to compute.

Corollary 2. *If $n_1 < n_2$ and both decks have identical composition, then $W_1 > \frac{1}{2}$: the player with the smaller combo has a strict advantage.*

8 Multiple Independent Combos

8.1 Motivation

A player may include multiple independent win conditions. For example, a deck might contain two separate combos $\mathcal{Z}^{(1)}$ and $\mathcal{Z}^{(2)}$ with no overlapping pieces. The player wins as soon as *any* combo is assembled.

Definition 7 (Independent Combos)**.** *Player i has C_i independent combos, $\mathcal{Z}_i^{(1)}, \ldots, \mathcal{Z}_i^{(C_i)}$, with pairwise disjoint card sets. The player's first assembly time is*

$$\tau_i = \min_{c=1,\ldots,C_i} \tau_i^{(c)},$$

where $\tau_i^{(c)}$ is the first turn on which combo c is complete.

8.2 Probability with Multiple Combos

Proposition 4 (At Least One Combo by Turn T)**.** *If the combos use disjoint card sets, the probability that Player i has assembled at least one combo by turn T is*

$$\Pr(\tau_i \leq T) = 1 - \Pr\left(\bigcap_{c=1}^{C_i} \{\tau_i^{(c)} > T\}\right). \tag{5}$$

For independent combos (disjoint card sets), the joint event requires that none *of the combos are complete. This is not the product of marginals (the draws are shared), so the computation requires the joint multivariate hypergeometric over all combo types simultaneously.*

Remark 3. *When C_i is small and the combos are disjoint, inclusion–exclusion is tractable:*

$$\Pr(\tau_i \leq T) = \sum_c P_{\mathcal{Z}_i^{(c)}}(T) - \sum_{c<c'} P_{\mathcal{Z}_i^{(c)} \cup \mathcal{Z}_i^{(c')}}(T) + \cdots$$

For general configurations, Monte Carlo is more practical, which we employ in Section 9.

8.3 Redundancy vs. Combo Size Trade-off

A key deck-building insight: a player with a larger combo ($n_2 > n_1$) can compensate by running more independent combos ($C_2 > C_1$). Intuitively, each additional combo path creates a new "lottery ticket" for assembling a win condition.

Theorem 2 (Redundancy Offset)**.** *Fix Player 1 with one combo of size n_1. There exists a finite C_2^* such that for $C_2 \geq C_2^*$, Player 2 (with combo size $n_2 > n_1$) has $W_2 > \frac{1}{2}$, provided $C_2 \cdot n_2 \leq X - K_L$ (the combos fit in the deck).*

Proof sketch. As C_2 increases, $\Pr(\tau_2 \leq T) \to 1$ for any $T \geq n_2$, while $\Pr(\tau_1 \leq T)$ remains fixed. Since Player 2's first-assembly distribution shifts leftward (stochastically dominates), there exists a threshold C_2^* at which W_2 exceeds $\frac{1}{2}$. The constraint $C_2 \cdot n_2 \leq X - K_L$ ensures the combos physically fit in the deck. □

9 Monte Carlo Simulations

All simulations are conducted in R. We use $B = 100{,}000$ replications per scenario unless otherwise noted.

9.1 Baseline: Combo Probability by Turn

Parameters: $X = 60$, $h = 7$, on the play. Deck composition: 24 lands, 15 creatures, 8 enchantments, 7 instants, 6 sorceries. We vary $N \in \{2, 3, 4, 5\}$ with the combo pieces distributed as: $\lfloor N/2 \rfloor$ creatures, $\lceil N/2 \rceil$ noncreature spells.

Figure 1 shows the cumulative probability of assembling the combo by turn T for each combo size. As expected, smaller combos are assembled faster. A 2-card combo reaches $> 50\%$ probability by turn 4–5, while a 5-card combo does not reach 50% until approximately turn 12–14.

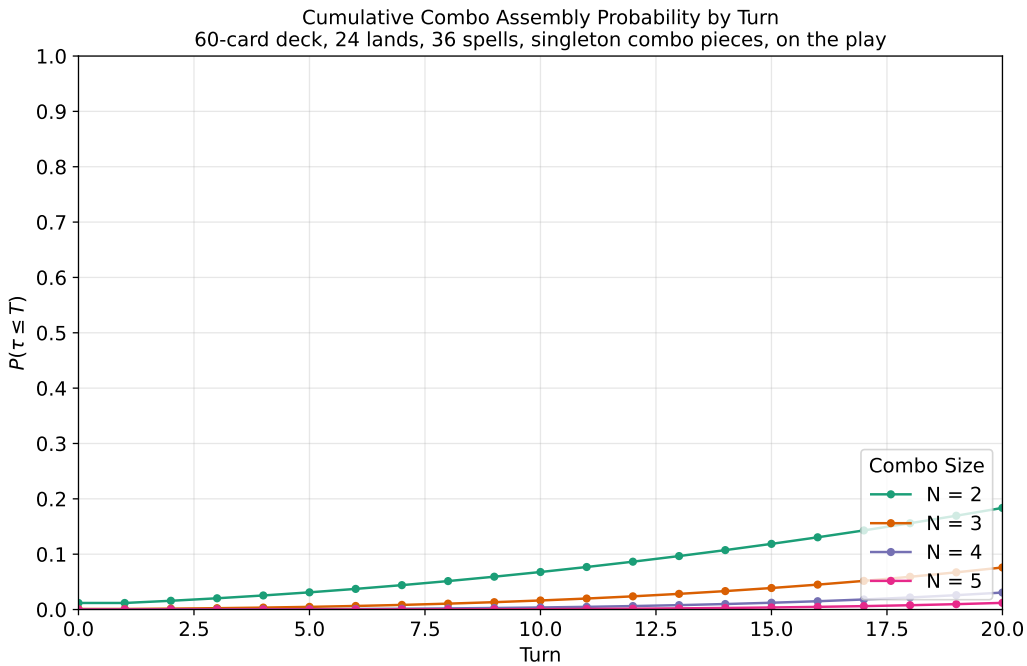

Figure 1: Cumulative probability of combo assembly by turn, for combo sizes $N = 2, 3, 4, 5$. Deck: 60 cards, 24 lands, 36 spells. Opening hand of 7, on the play.

9.2 Deck Composition Sensitivity

We fix $N = 3$ (requiring 1 creature and 2 instants) and vary the number of instants in the deck from 4 to 20, holding $X = 60$ and lands at 24 (adjusting creatures to fill the remaining slots).

Figure 2 shows that increasing the count of the relevant card type (instants) substantially raises the combo probability at every turn, illustrating Proposition 2.

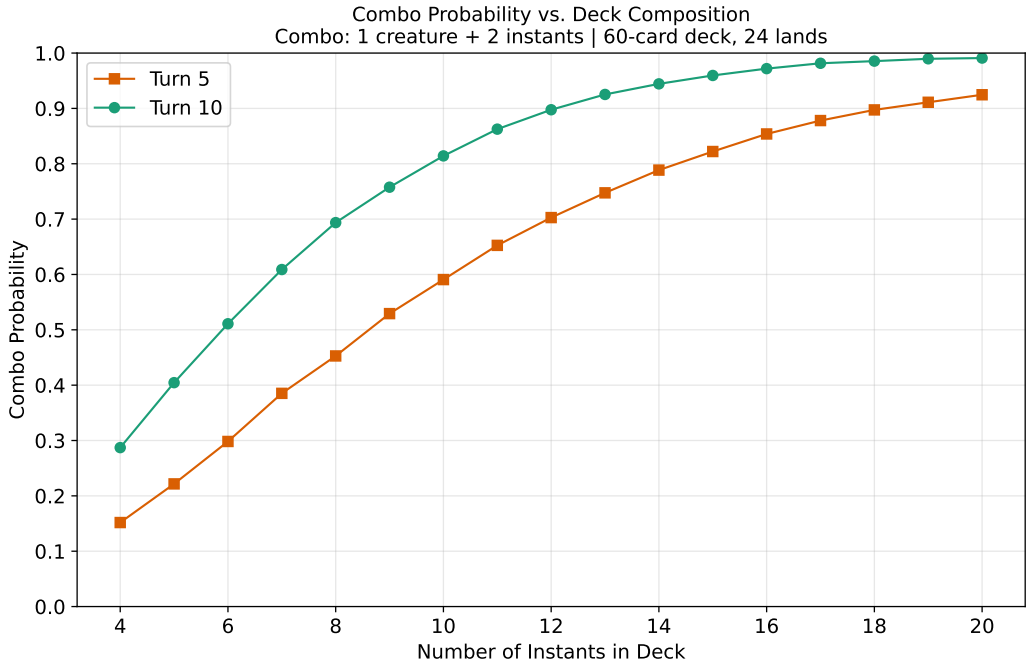

Figure 2: Combo probability at turn 5 and turn 10 as a function of the number of instants in the deck. Combo: 1 creature + 2 instants.

9.3 Mulligan Effect

Figure 3 compares the combo probability for $N = 3$ under no mulligan ($m = 0$), one mulligan ($m = 1$, greedy policy), and two mulligans ($m = 2$). The greedy mulligan policy offers a second chance at the opening hand but reduces effective hand size, creating a trade-off that depends on combo size relative to hand size.

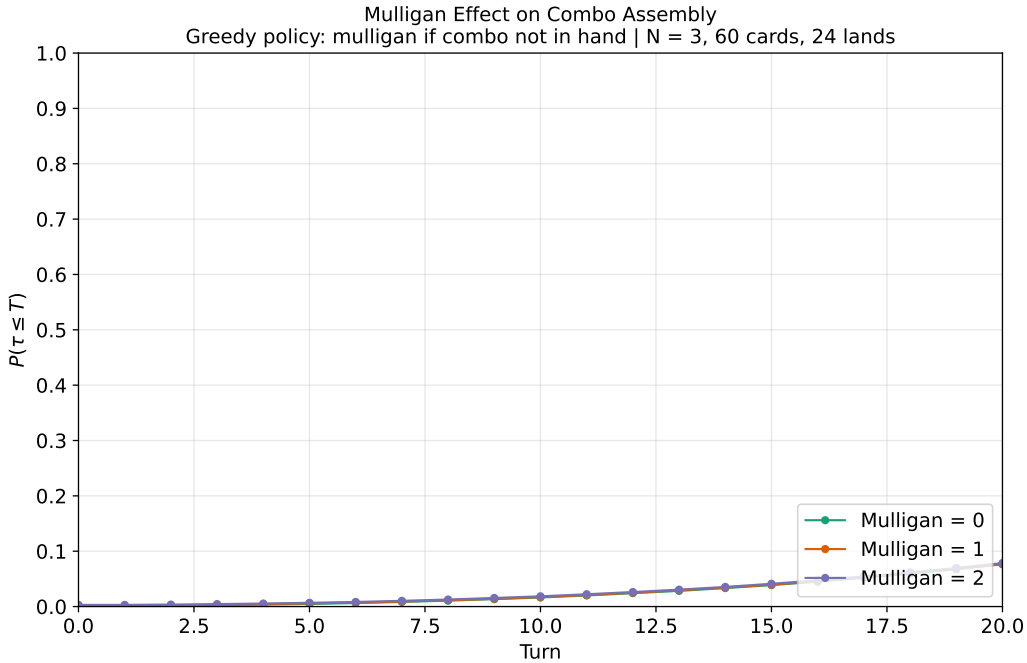

Figure 3: Combo probability by turn under different mulligan counts. Greedy policy: mulligan if combo not in opening hand. $N = 3$, 60-card deck, 24 lands.

9.4 Two-Player Race

We simulate the two-player race with $n_1 = 2$ vs. $n_2 \in \{3, 4, 5\}$. Both players have 60-card decks with identical composition (24 lands, 36 spells). Combo pieces are singletons drawn from the spell portion.

Figure 4 shows Player 1's win probability as a function of n_2. The advantage to the smaller combo is substantial: with $n_1 = 2$ vs. $n_2 = 5$, Player 1 wins roughly 75–80% of the time.

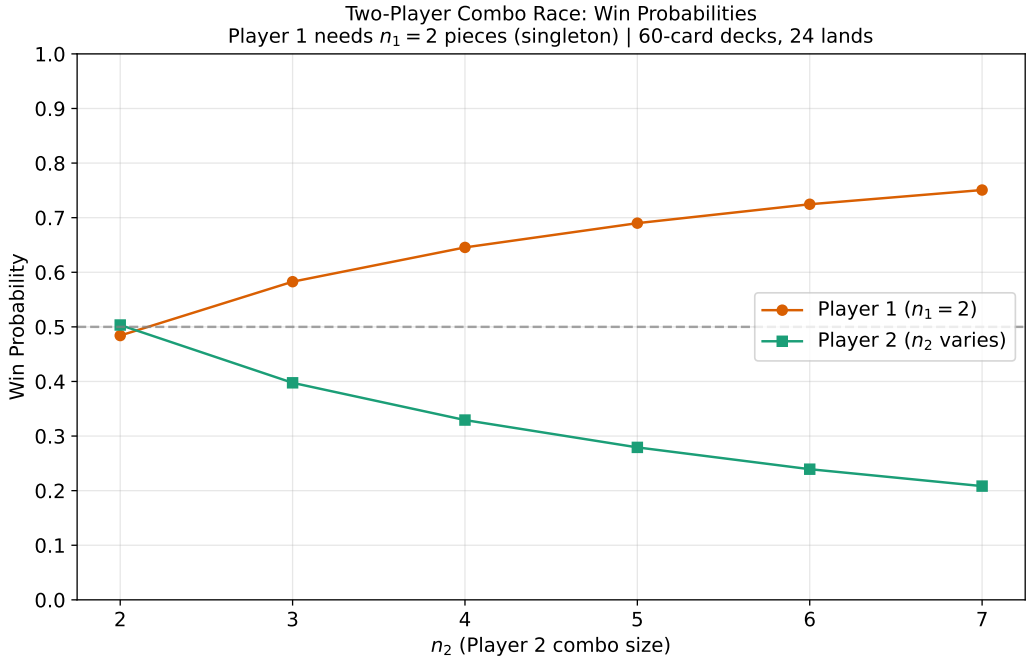

Figure 4: Player 1 win probability in the combo race. Player 1 needs $n_1 = 2$ pieces; Player 2 needs n_2 pieces. Both decks: 60 cards, 24 lands, 36 spells, singleton combo pieces.

9.5 Multiple Combos: Redundancy Effect

We fix Player 1 at $n_1 = 2$ with $C_1 = 1$ combo. Player 2 has $n_2 = 4$ but varies $C_2 \in \{1, 2, 3, 4\}$ independent combos (each requiring 4 distinct singletons).

Figure 5 demonstrates the redundancy offset from Theorem 2: as Player 2 adds more independent combos, their win probability rises, eventually overcoming the combo-size disadvantage.

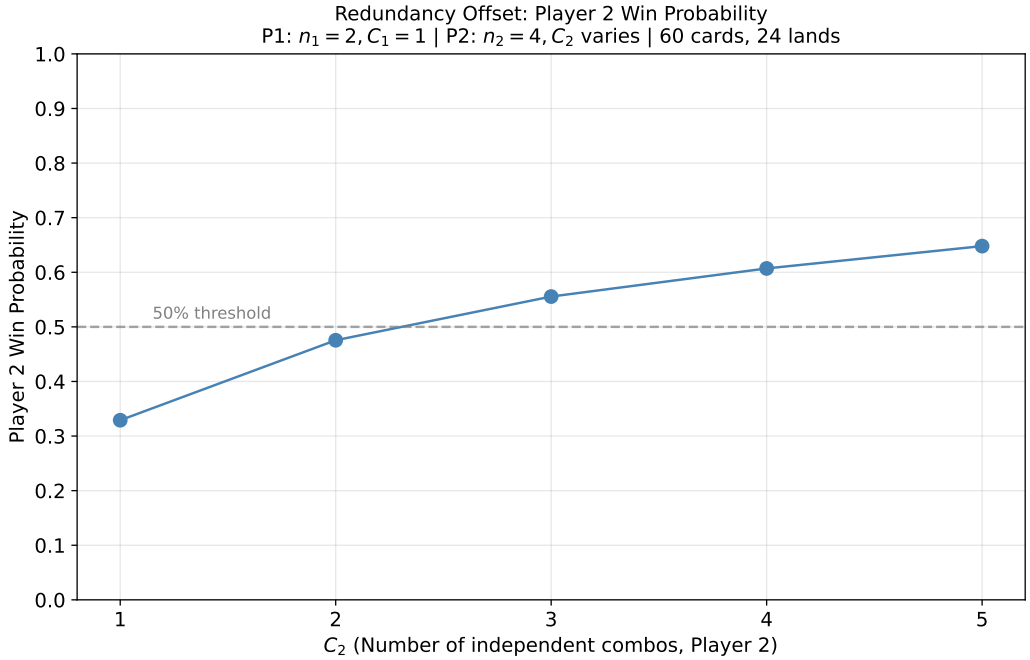

Figure 5: Player 2 win probability as a function of the number of independent combos C_2. Player 1: $n_1 = 2$, $C_1 = 1$. Player 2: $n_2 = 4$. Both decks: 60 cards, 24 lands.

9.6 Heatmap: Combo Size vs. Number of Combos

Figure 6 presents a heatmap of Player 2's win probability across a grid of (n_2, C_2) values, with Player 1 fixed at $(n_1 = 2, C_1 = 1)$. The iso-probability contours show the substitution rate between combo size and redundancy.

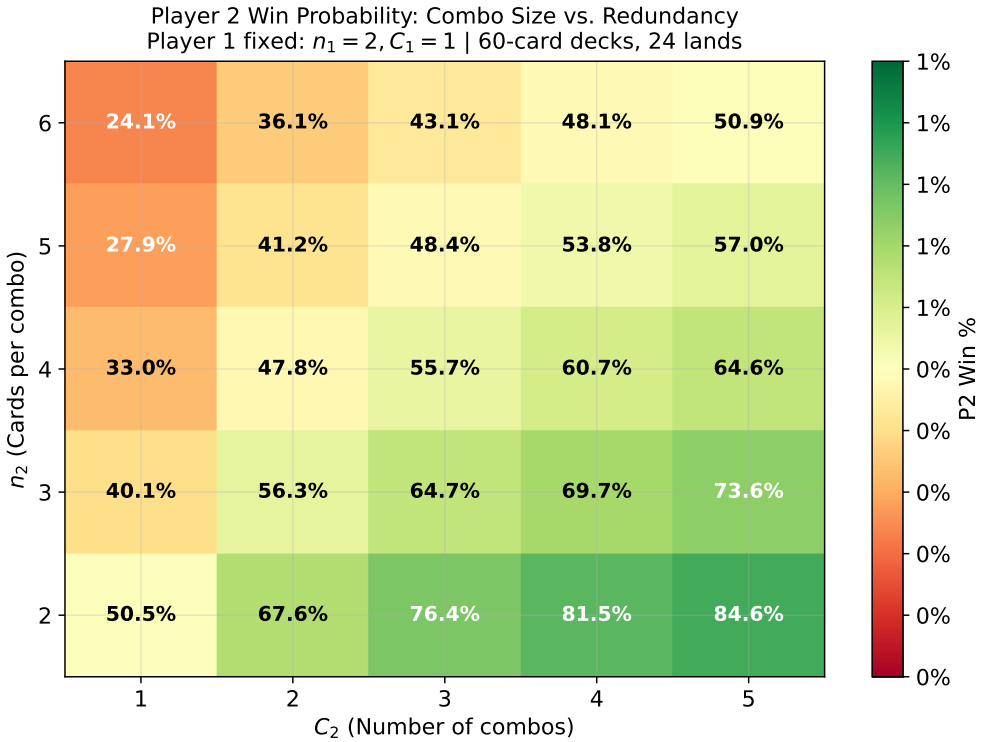

Figure 6: Heatmap of Player 2 win probability over combo size n_2 and number of independent combos C_2. Player 1: $n_1 = 2$, $C_1 = 1$.

10 Discussion

The framework developed here provides a rigorous foundation for analyzing combo-based strategies in constructed card games. Several points merit discussion.

Practical deck building. The model quantifies the intuition that smaller combos are more reliable. A 2-card combo in a 60-card deck with adequate redundancy (e.g., 4 copies of each piece) is assembled far more quickly than a 5-card combo, even with maximal redundancy. The two-player race results formalize the competitive advantage of lean combo packages.

The redundancy–complexity trade-off. Theorem 2 and the simulations demonstrate that a player can compensate for a larger combo requirement by running multiple independent win conditions. This mirrors a well-known principle in competitive MTG: "toolbox" decks with multiple lines of attack are more resilient than single-combo decks with complex requirements.

Interaction effects. The model treats combo assembly as a pure draw problem and ignores player interaction (counterspells, discard, removal). In practice, the opponent can disrupt combo assembly, which would require extending the model to include "removal" shocks that decrease the effective cards seen. This is a natural direction for future work.

Mana constraints. Drawing combo pieces is necessary but not sufficient—the player must also have the mana to cast them. Incorporating mana requirements transforms the

problem into one with dependent constraints (lands and spells must both be present in sufficient quantities), which is handled by the multivariate hypergeometric framework but adds complexity to the optimization.

Bayesian updating. As the game progresses, each player gains information about their remaining library. The conditional probability of assembling the combo given cards already seen updates according to a hypergeometric posterior. This dynamic is captured implicitly in our turn-by-turn framework but could be made explicit in a Bayesian formulation.

11 Conclusion

We have developed a probabilistic framework for combo assembly in constructed card games, grounded in the multivariate hypergeometric distribution. The model accommodates arbitrary deck sizes, card-type partitions, mulligan policies, and extra draw effects. The two-player race extension and the multiple-combo redundancy theorem provide formal tools for evaluating deck-building decisions in competitive settings.

The key quantitative results are: (i) combo probability is well-approximated by the product formula in (3) for singleton combos; (ii) deck composition has a first-order effect on assembly speed, with the optimal type allocation exhibiting an interior maximum; (iii) in head-to-head races, the player with the smaller combo enjoys a significant advantage that can only be offset by substantial redundancy in the opponent's deck.

The R simulation code accompanying this paper is provided in the appendix and can be used to evaluate any specific deck configuration and combo requirement.

A R Simulation Code

The complete R code for all simulations and figures is provided in the accompanying file `simulations.R`. Key functions:

- `combo_prob_exact(N, X, d)`: Computes exact combo probability using (3).
- `combo_prob_mv_hyper(n_vec, K_vec, X, d)`: Multivariate hypergeometric combo probability via Monte Carlo.
- `simulate_race(n1, n2, X, K_L, B)`: Simulates the two-player combo race.
- `simulate_multi_combo(n1, C1, n2, C2, X, K_L, B)`: Simulates the multi-combo race.